Build Task — Project 2: Tool-Using Agent — Hints & Solution

> Back to the Build Task · Hint 1 · Hint 2 · Solution

Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried.

- Hint 1 — What you're building, and the exact pieces
- Hint 2 — The plan, and almost the whole thing
- Solution

> Back to the Build Task · Hint 1 · Hint 2 · Solution

Hint 1 — What you're building, and the exact pieces

The one new piece is a tool that **fails once, then works**, so the recovery Test Case is repeatable:

- `docs_search.py`: a `FlakyToolState` class with a `call_count`, one shared `search_state = FlakyToolState()`, and `@tool search_docs(query)` that raises `ConnectionError` when `call_count == 1`.
- A test sets `search_state.call_count = 0` before the run, so the first search really fails.

`MaxIterationsExceeded` moves into `exceptions.py`, next to Doc02's named errors. `test_agent.py` has one function per Test Cases row, in the plain `case` print style of Doc04–06 — or as pytest tests, each with a fresh fail-once tool from the Intermediate exercise's factory. For your own argument checks, reuse its `run_tool_checked()` idea: an `ARG_MODELS` dict and `model_dump()`.

> Back to the Build Task · Hint 1 · Hint 2 · Solution

Hint 2 — The plan, and almost the whole thing

```
# docs_search.py — the new piece
class FlakyToolState:
```

```
    def __init__(self) -> None:
        self.call_count = 0

search_state = FlakyToolState()

@tool
def search_docs(query: str) -> str:
    """Search the company's policy documents (refunds, shipping)."""
    search_state.call_count = search_state.call_count + 1
    if search_state.call_count == 1:
        raise ConnectionError("search index temporarily unavailable")
    ...
```

`agent.py` is the Intermediate exercise's loop, now with `AGENT_TOOLS` instead of `ALL_TOOLS`. Finish the files, then compare against the Solution.

> Back to the Build Task · Hint 1 · Hint 2 · Solution

Solution

Story — `exceptions.py`: Doc06's file, unchanged, plus `MaxIterationsExceeded` — the same named-error pattern as `PermanentHTTPError`, carrying the partial step log. **If not:** a run that hit the limit would look the same as a crash.

```
# exceptions.py — the class added at the end of Doc06's file
class MaxIterationsExceeded(Exception):
    """Raised when the agent loop hits its step limit, with the log."""

    def __init__(self, message: str, steps: list[dict]) -> None:
        # why: the partial step log travels with the error, so the caller
        # can see exactly what happened before it gave up
        self.steps = steps
        super().__init__(message)
```

Story — `docs_search.py`: a tool that fails on its first call, on purpose, controlled by a counter a test can reset. **If not:** tool failures would only happen by bad luck, and recovery could never be tested on demand.

```
# docs_search.py
from langchain_core.tools import tool

# why: a small, known "document store" — a test can check the agent
```

```
# really used it instead of guessing
COMPANY_DOCS = {
    "refund": "Refunds are accepted within 30 days, with the receipt.",
    "shipping": "Standard shipping takes 3-5 working days.",
}

class FlakyToolState:
    def __init__(self) -> None:
        self.call_count = 0

# how: one shared counter; a test sets it back to 0 before each run
search_state = FlakyToolState()

@tool
def search_docs(query: str) -> str:
    """Search the company's policy documents (refunds, shipping)."""
    search_state.call_count = search_state.call_count + 1
    if search_state.call_count == 1:
        # why: fails on the first call on purpose, so a run proves the
        # agent recovers instead of crashing or making up an answer
        raise ConnectionError("search index temporarily unavailable")
    lowered = query.lower()
    for topic in COMPANY_DOCS:
        if topic in lowered:
            return COMPANY_DOCS[topic]
    return "No matching document found."
```

Story — `agent.py` : the Intermediate exercise's loop, now with `AGENT_TOOLS` — two Doc06 tools and the fail-once search — and the same optional `tools` list, so a test can pass its own. **If not:** Project 2 would be the first place these pieces ever ran together.

```
# agent.py
from dataclasses import dataclass, field

from langchain_core.tools import BaseTool
from langchain_openai import ChatOpenAI

from docs_search import search_docs
from exceptions import MaxIterationsExceeded
from logging_setup import get_logger
from tools import add, get_weather

# why: Doc06's tools, reused unchanged, plus the one that fails once
AGENT_TOOLS = [add, get_weather, search_docs]

@dataclass
```

```
class AgentResult:
    final_answer: str
    # why: a list default needs field(default_factory=list), not `= []`
    steps: list[dict] = field(default_factory=list)

def find_tool(tools: list[BaseTool], name: str) -> BaseTool | None:
    # how: Doc06's Build Task function, copied unchanged
    for candidate in tools:
        if candidate.name == name:
            return candidate
    return None

def run_tool(
    call: dict, tools: list[BaseTool]
) -> tuple[str | None, str | None]:
    # how: returns (result, None) on success, (None, error) on failure
    tool = find_tool(tools, call["name"])
    if tool is None:
        return None, f"unknown tool: {call['name']}"
    try:
        # how: .invoke() checks the arguments with the tool's Pydantic
        # args_schema first (Doc06) — bad arguments never reach the tool
        return str(tool.invoke(call["args"])), None
    except Exception as e:
        # why: any tool failure becomes an Observation, never a crash
        return None, str(e)

def run_agent(
    task: str, max_iterations: int, tools: list[BaseTool] | None = None
) -> AgentResult:
    # why: None means "use AGENT_TOOLS"; a test can pass its own tools
    if tools is None:
        tools = AGENT_TOOLS
    logger = get_logger(__name__)
    model = ChatOpenAI(model="gpt-4o-mini", temperature=0)
    model_with_tools = model.bind_tools(tools)
    messages = [{"role": "user", "content": task}]
    steps: list[dict] = []

    for step in range(max_iterations):
        response = model_with_tools.invoke(messages)
        messages.append(response)
        if not response.tool_calls:
            return AgentResult(final_answer=response.content, steps=steps)

        # why: run EVERY call the model asked for — each needs an answer
        for call in response.tool_calls:
```

```
            result, error = run_tool(call, tools)
            if error is None:
                observation = result
                logger.info(f"step {step}: {call['name']} -> {result}")
            else:
                observation = f"Error: {error}"
                # how: a warning, not an error — the loop keeps going
                logger.warning(f"step {step}: {call['name']} failed: {error}")
            messages.append({
                "role": "tool",
                "content": observation,
                "tool_call_id": call["id"],
            })
            steps.append({
                "step": step,
                "tool": call["name"],
                "arguments": call["args"],
                "result": result,
                "error": error,
            })

    message = f"No answer after {max_iterations} steps"
    raise MaxIterationsExceeded(message, steps)
```

Story — `main.py`: reads a task, runs it, and prints the answer and every step — or, when the limit is hit, the partial log. **If not:** a stopped run would end in a traceback instead of an explanation.

```
# main.py
from agent import run_agent
from exceptions import MaxIterationsExceeded

MAX_ITERATIONS = 5

def main() -> None:
    task = input("What do you need? ")
    try:
        result = run_agent(task, MAX_ITERATIONS)
    except MaxIterationsExceeded as e:
        # why: even a failed run shows its full step log
        print(f"Gave up: {e}")
        for step in e.steps:
            print(step)
        return

    print(f"Answer: {result.final_answer}")
    for step in result.steps:
```

```
        print(step)

if __name__ == "__main__":
    main()
```

Story — `test_agent.py` : the 4 Test Cases, one function each, in the plain `case` style of Doc04–06. **If not:** "it recovers" and "it stops" would be things you checked once by hand.

```
# test_agent.py
# Runs the 4 Test Cases from the README. Run it from inside this folder:
# python test_agent.py
# Makes real API calls — needs a valid OPENAI_API_KEY in .env.
from agent import run_agent
from docs_search import search_state
from exceptions import MaxIterationsExceeded

def test_clear_task_uses_the_tool() -> None:
    result = run_agent("What's 5 + 7?", 5)
    tools_used = []
    for step in result.steps:
        tools_used.append(step["tool"])
    print(f"case 1 (clear task)   -> tools: {tools_used}")
    print(f"                         answer: {result.final_answer}")

def test_no_tool_needed() -> None:
    result = run_agent("What's the capital of France?", 5)
    print(f"case 2 (no tool)      -> {len(result.steps)} tool calls")

def test_recovers_from_a_failure() -> None:
    # how: reset the shared counter, so the first search really fails
    search_state.call_count = 0
    result = run_agent("What is the refund policy?", 5)
    failed = 0
    succeeded = 0
    for step in result.steps:
        if step["error"] is None:
            succeeded = succeeded + 1
        else:
            failed = failed + 1
    print(f"case 3 (recovery)     -> failed={failed}, succeeded={succeeded}")
    print(f"                         answer: {result.final_answer}")

def test_step_limit() -> None:
    # how: a two-step task with room for only one step
    try:
        run_agent("Add 1 + 2, then add 3 to that.", 1)
```

```
        print("case 4 (step limit)   -> no error (wrong!)")
    except MaxIterationsExceeded as e:
        print(f"case 4 (step limit)   -> raised, {len(e.steps)} step logged")

if __name__ == "__main__":
    test_clear_task_uses_the_tool()
    test_no_tool_needed()
    test_recovers_from_a_failure()
    test_step_limit()
```

Expected output (answers vary; one `WARNING` log line appears for the failed search):

```
case 1 (clear task)   -> tools: ['add']
                         answer: 5 + 7 is 12.
case 2 (no tool)      -> 0 tool calls
case 3 (recovery)     -> failed=1, succeeded=1
                         answer: Refunds are accepted within 30 days.
case 4 (step limit)   -> raised, 1 step logged
```

> Back to the Build Task · Hint 1 · Hint 2 · Solution